

Linear Functions & Inequalities

ANSWER KEY



Slope and slope-intercept form

- 1 Find the slope of the line passing through (2, 3) and (6, 11).

$$\text{Slope} = \frac{11 - 3}{6 - 2} = \frac{8}{4} = 2.$$

- 2 Write the equation of the line with slope -3 passing through (1, 4), in slope-intercept form.

$$y - 4 = -3(x - 1), \text{ so } y = -3x + 3 + 4 = -3x + 7.$$

Parallel and perpendicular lines

- 3 Find the equation of the line parallel to $y = 2x - 5$ passing through (3, 4).

Parallel lines share the same slope, $m = 2$. $y - 4 = 2(x - 3)$, so $y = 2x - 6 + 4 = 2x - 2$.

- 4 Find the equation of the line perpendicular to $y = \frac{1}{2}x + 3$ passing through (4, 1).

The perpendicular slope is the negative reciprocal: $m = -2$. $y - 1 = -2(x - 4)$, so $y = -2x + 8 + 1 = -2x + 9$.

Finding equations from two points or a table

- 5 A linear function passes through (0, 5) and (4, 17). Find its equation.

$$\text{Slope} = \frac{17 - 5}{4 - 0} = \frac{12}{4} = 3. \text{ Since } (0, 5) \text{ is the } y\text{-intercept, the equation is } y = 3x + 5.$$

- 6 A table shows $x = 1, y = 7$ and $x = 3, y = 15$ for a linear function. Find the equation.

$$\text{Slope} = \frac{15 - 7}{3 - 1} = \frac{8}{2} = 4. \text{ Using point } (1, 7): y - 7 = 4(x - 1), \text{ so } y = 4x - 4 + 7 = 4x + 3.$$

Solving and graphing linear inequalities

- 7 Solve the inequality $3x - 4 < 11$, and graph the solution on a number line description.

$3x < 15$, so $x < 5$. On a number line, this is an open circle at 5 with shading extending to the left (all values less than 5).

- 8 Solve the inequality $-2x + 7 \geq 1$, being careful with the direction of the inequality sign.

$$-2x \geq -6. \text{ Dividing by } -2 \text{ flips the inequality: } x \leq 3.$$

Systems of linear inequalities

- 9 A system requires $y > 2x - 1$ and $y \leq -x + 5$. Determine whether the point $(1, 2)$ satisfies both inequalities.

Check the first: $2 > 2(1) - 1 = 1$? True. Check the second: $2 \leq -1 + 5 = 4$? True. Since $(1, 2)$ satisfies both inequalities, it lies in the solution region.

Interpreting linear inequalities in context

- 10 A student has 50 dollars and wants to buy notebooks costing 6 dollars each, keeping at least 8 dollars in reserve. Write an inequality for the number of notebooks n the student can buy, and solve it.

The amount spent, $6n$, must leave at least 8 dollars: $50 - 6n \geq 8$, so $-6n \geq -42$, giving $n \leq 7$ (dividing by a negative number flips the inequality). The student can buy at most 7 notebooks.

- 11 A factory produces at least 200 units per day but at most 350 units per day. Write a compound inequality for the daily production p , and state whether $p = 180$ is a valid production level.

$200 \leq p \leq 350$. Since $180 < 200$, the value $p = 180$ does NOT satisfy the inequality — it is below the minimum required production level.

Comparing rates of change between two linear functions

- 12 Function f is given by $f(x) = 3x + 2$. Function g is defined by the table: $x = 0, g = 1$ and $x = 2, g = 9$. Which function has the greater rate of change?

f has a rate of change (slope) of 3. For g : slope $= \frac{9 - 1}{2 - 0} = \frac{8}{2} = 4$. Since $4 > 3$, function g has the greater rate of change.

Finding the intersection of two lines graphically and algebraically

- 13 Two lines are given by $y = 2x - 1$ and $y = -x + 8$. Find their point of intersection, and explain what this point represents graphically.

Set equal: $2x - 1 = -x + 8$, so $3x = 9$, giving $x = 3$. Then $y = 2(3) - 1 = 5$. The lines intersect at $(3, 5)$, which represents the single point where both lines cross on a graph — the unique solution to the system formed by the two equations.

Absolute value equations and inequalities

14 Solve $|2x - 3| = 9$.

Either $2x - 3 = 9$, giving $x = 6$, or $2x - 3 = -9$, giving $x = -3$. Solutions: $x = 6$ or $x = -3$.

15 Solve $|x + 4| < 6$, giving your answer as an interval.

$-6 < x + 4 < 6$, so $-10 < x < 2$.

Writing a linear inequality from a verbal description

- 16 A shipping company charges 3 dollars per pound but offers free shipping if the package weighs less than 2 pounds. Write a piecewise description of the cost $C(w)$ as a function of weight w , and find the cost of a 5-pound package.

$C(w) = 0$ for $w < 2$ and $C(w) = 3w$ for $w \geq 2$. For a 5-pound package: $C(5) = 3(5) = 15$ dollars.

- 17 A parking garage charges 5 dollars for the first hour and 2 dollars for each additional hour. Write an equation for the total cost C after h hours (for $h \geq 1$), and find the cost for 6 hours.

$C(h) = 5 + 2(h - 1)$ for $h \geq 1$ (since only the hours after the first are charged at the additional rate). For 6 hours: $C(6) = 5 + 2(5) = 5 + 10 = 15$ dollars.

Graphing a system of inequalities and identifying the feasible region

- 18 A system requires $x + y \leq 10$ and $x \geq 2$ and $y \geq 0$. Determine which of the points $(3, 4)$ and $(8, 5)$ lie in the feasible (solution) region, checking each condition.

For $(3, 4)$: $3 + 4 = 7 \leq 10$ ✓, $3 \geq 2$ ✓, $4 \geq 0$ ✓ — all conditions satisfied, so $(3, 4)$ is in the feasible region. For $(8, 5)$: $8 + 5 = 13 \leq 10$? False — this point fails the first condition, so $(8, 5)$ is NOT in the feasible region.

Rate of change from a real-world linear model

- 19 A water tank starts with 500 gallons and drains at a constant rate. After 4 hours, it contains 340 gallons. Write a linear equation for the volume V after t hours, and predict when the tank will be empty.

Rate of change: $\frac{340 - 500}{4 - 0} = \frac{-160}{4} = -40$ gallons per hour. Equation: $V = 500 - 40t$. The tank is empty when $V = 0$: $500 - 40t = 0$, so $t = 12.5$ hours.

Multiple choice — easy 15

20 Find the slope of the line passing through $(1, -4)$ and $(8, -6)$.

- a. $-\frac{7}{2}$
- b. $-\frac{2}{7}$ ✓
- c. $-\frac{9}{7}$
- d. $\frac{5}{7}$

$$\text{Slope} = (-6 - (-4))/(8 - (1)) = -2/7 = -\frac{2}{7}.$$

21 Find the slope of the line passing through $(3, 5)$ and $(8, -2)$.

- a. $-\frac{5}{7}$
- b. $-\frac{7}{5}$ ✓
- c. $-\frac{12}{5}$
- d. $-\frac{2}{5}$

$$\text{Slope} = (-2 - (5))/(8 - (3)) = -7/5 = -\frac{7}{5}.$$

22 Find the slope of the line passing through $(-5, -9)$ and $(6, 1)$.

- a. $\frac{11}{10}$
- b. $-\frac{1}{11}$
- c. $\frac{10}{11}$ ✓
- d. $\frac{21}{11}$

$$\text{Slope} = (1 - (-9))/(6 - (-5)) = 10/11 = \frac{10}{11}.$$

23 Find the slope of the line passing through $(4, 5)$ and $(5, -9)$.

- a. -14 ✓
- b. $-\frac{1}{14}$
- c. -15
- d. -13

$$\text{Slope} = (-9 - (5))/(5 - (4)) = -14/1 = -14.$$

24 Find the slope of the line passing through $(3, -10)$ and $(-7, 8)$.

a. $-\frac{14}{5}$

b. $-\frac{9}{5}$ ✓

c. $-\frac{5}{9}$

d. $-\frac{4}{5}$

$$\text{Slope} = (8 - (-10))/(-7 - (3)) = 18/-10 = -\frac{9}{5}.$$

25 Find the slope of the line passing through $(-2, 5)$ and $(3, -3)$.

a. $-\frac{13}{5}$

b. $-\frac{5}{8}$

c. $-\frac{8}{5}$ ✓

d. $-\frac{3}{5}$

$$\text{Slope} = (-3 - (5))/(3 - (-2)) = -8/5 = -\frac{8}{5}.$$

26 Find the slope of the line passing through $(-4, -9)$ and $(6, 0)$.

a. $\frac{9}{10}$ ✓

b. $-\frac{1}{10}$

c. $\frac{10}{9}$

d. $\frac{19}{10}$

$$\text{Slope} = (0 - (-9))/(6 - (-4)) = 9/10 = \frac{9}{10}.$$

27 Find the slope of the line passing through $(9, -8)$ and $(-2, 9)$.

a. $-\frac{17}{11}$ ✓

b. $-\frac{28}{11}$

c. $-\frac{11}{17}$

d. $-\frac{6}{11}$

$$\text{Slope} = (9 - (-8))/(-2 - (9)) = 17/-11 = -\frac{17}{11}.$$

28 Find the slope of the line passing through $(-7, 9)$ and $(1, 8)$.

a. $-\frac{1}{8}$ ✓

b. -8

c. $\frac{7}{8}$

d. $-\frac{9}{8}$

$$\text{Slope} = (8 - (9))/(1 - (-7)) = -1/8 = -\frac{1}{8}.$$

29 Find the slope of the line passing through $(-7, -7)$ and $(7, 8)$.

a. $\frac{29}{14}$

b. $\frac{14}{15}$

c. $\frac{1}{14}$

d. $\frac{15}{14}$ ✓

$$\text{Slope} = (8 - (-7))/(7 - (-7)) = 15/14 = \frac{15}{14}.$$

30 Find the slope of the line passing through $(2, -3)$ and $(-1, 1)$.

a. $-\frac{7}{3}$

b. $-\frac{1}{3}$

c. $-\frac{4}{3}$ ✓

d. $-\frac{3}{4}$

$$\text{Slope} = (1 - (-3))/(-1 - (2)) = 4/-3 = -\frac{4}{3}.$$

31 Find the slope of the line passing through (8, 6) and (−5, − 7).

a. 1 ✓

b. 3

c. 0

d. 2

$$\text{Slope} = (-7 - (6))/(-5 - (8)) = -13/-13 = 1.$$

32 Find the slope of the line passing through (−3, 3) and (5, − 10).

a. $-\frac{5}{8}$

b. $-\frac{13}{8}$ ✓

c. $-\frac{8}{13}$

d. $-\frac{21}{8}$

$$\text{Slope} = (-10 - (3))/(5 - (-3)) = -13/8 = -\frac{13}{8}.$$

33 Find the slope of the line passing through (0, − 2) and (8, − 3).

a. $\frac{7}{8}$

b. −8

c. $-\frac{9}{8}$

d. $-\frac{1}{8}$ ✓

$$\text{Slope} = (-3 - (-2))/(8 - (0)) = -1/8 = -\frac{1}{8}.$$

34 Find the slope of the line passing through $(-3, 8)$ and $(3, -6)$.

- a. $-\frac{4}{3}$
- b. $-\frac{10}{3}$
- c. $-\frac{7}{3}$ ✓
- d. $-\frac{3}{7}$

$$\text{Slope} = (-6 - 8)/(3 - (-3)) = -14/6 = -\frac{7}{3}.$$

Multiple choice — medium 15

35 Solve the inequality $-x - 17 \leq -21$.

- a. $x > 4$
- b. $x < 4$
- c. $x \leq 4$
- d. $x \geq 4$ ✓

$-x \leq -4$. Dividing by a negative number flips the inequality. $x \geq 4$.

36 Solve the inequality $5x + 19 \leq -31$.

- a. $x < -10$
- b. $x > -10$
- c. $x \geq -10$
- d. $x \leq -10$ ✓

$$5x \leq -50. x \leq -10.$$

37 Solve the inequality $-5x < 45$.

a. $x > -9$ ✓

b. $x < -9$

c. $x \geq -9$

d. $x \leq -9$

$-5x < 45$. Dividing by a negative number flips the inequality. $x > -9$.

38 Solve the inequality $-2x - 3 \leq -17$.

a. $x \leq 7$

b. $x > 7$

c. $x \geq 7$ ✓

d. $x < 7$

$-2x \leq -14$. Dividing by a negative number flips the inequality. $x \geq 7$.

39 Solve the inequality $x + 20 < 34$.

a. $x < 14$ ✓

b. $x > 14$

c. $x \leq 14$

d. $x \geq 14$

$x < 14$.

40 Solve the inequality $7x + 16 \leq 93$.

a. $x \geq 11$

b. $x \leq 11$ ✓

c. $x > 11$

d. $x < 11$

$7x \leq 77$. $x \leq 11$.

41 Solve the inequality $-8x - 20 > -132$.

a. $x \leq 14$

b. $x < 14$ ✓

c. $x > 14$

d. $x \geq 14$

$-8x > -112$. Dividing by a negative number flips the inequality. $x < 14$.

42 Solve the inequality $-5x - 5 < -45$.

a. $x \leq 8$

b. $x > 8$ ✓

c. $x < 8$

d. $x \geq 8$

$-5x < -40$. Dividing by a negative number flips the inequality. $x > 8$.

43 Solve the inequality $x + 15 \leq 18$.

a. $x \geq 3$

b. $x \leq 3$ ✓

c. $x < 3$

d. $x > 3$

$x \leq 3$.

44 Solve the inequality $5x + 3 \leq 18$.

a. $x \geq 3$

b. $x > 3$

c. $x < 3$

d. $x \leq 3$ ✓

$5x \leq 15$. $x \leq 3$.

45 Solve the inequality $4x - 7 \leq -23$.

a. $x \leq -4$ ✓

b. $x > -4$

c. $x < -4$

d. $x \geq -4$

$4x \leq -16$. $x \leq -4$.

46 Solve the inequality $-3x + 17 > 5$.

a. $x \leq 4$

b. $x < 4$ ✓

c. $x \geq 4$

d. $x > 4$

$-3x > -12$. Dividing by a negative number flips the inequality. $x < 4$.

47 Solve the inequality $3x + 11 \leq 56$.

a. $x > 15$

b. $x \leq 15$ ✓

c. $x \geq 15$

d. $x < 15$

$3x \leq 45$. $x \leq 15$.

48 Solve the inequality $x + 10 \leq 2$.

a. $x < -8$

b. $x \leq -8$ ✓

c. $x \geq -8$

d. $x > -8$

$x \leq -8$.

49 Solve the inequality $-2x - 6 < -22$.

a. $x < 8$

b. $x > 8$ ✓

c. $x \geq 8$

d. $x \leq 8$

$-2x < -16$. Dividing by a negative number flips the inequality. $x > 8$.

Multiple choice — hard 15

50 Find the equation of the line perpendicular to $y = -\frac{2}{3}x - 6$ passing through $(5, 3)$, in slope-intercept form.

a. $y = -\frac{2}{3}x - \frac{9}{2}$

b. $y = \frac{3}{2}x - \frac{7}{2}$

c. $y = \frac{3}{2}x - \frac{9}{2}$ ✓

d. $y = \frac{3}{2}x + \frac{9}{2}$

The perpendicular slope is the negative reciprocal, $m = \frac{3}{2}$. $y - (3) = \frac{3}{2}(x - (5))$, so $y = \frac{3}{2}x - \frac{9}{2}$.

- 51 Find the equation of the line perpendicular to $y = x + 3$ passing through $(-4, -8)$, in slope-intercept form.

a. $y = x - 12$

b. $y = -x - 12$ ✓

c. $y = -x + 12$

d. $y = -x - 11$

The perpendicular slope is the negative reciprocal, $m = -1$. $y - (-8) = -1(x - (-4))$, so $y = -x - 12$.

- 52 Find the equation of the line parallel to $y = -3x + 1$ passing through $(4, -3)$, in slope-intercept form.

a. $y = -3x + 9$ ✓

b. $y = -3x - 9$

c. $y = -3x + 10$

d. $y = 3x + 9$

Parallel lines share the same slope, $m = -3$. $y - (-3) = -3(x - (4))$, so $y = -3x + 9$.

- 53 Find the equation of the line parallel to $y = \frac{1}{2}x - 5$ passing through $(4, -3)$, in slope-intercept form.

a. $y = \frac{1}{2}x + 5$

b. $y = \frac{1}{2}x - 5$ ✓

c. $y = \frac{1}{2}x - 4$

d. $y = -\frac{1}{2}x - 5$

Parallel lines share the same slope, $m = \frac{1}{2}$. $y - (-3) = \frac{1}{2}(x - (4))$, so $y = \frac{1}{2}x - 5$.

54 Find the equation of the line parallel to $y = \frac{1}{3}x - 7$ passing through $(2, 8)$, in slope-intercept form.

a. $y = \frac{1}{3}x + \frac{25}{3}$

b. $y = -\frac{1}{3}x + \frac{22}{3}$

c. $y = \frac{1}{3}x + \frac{22}{3}$ ✓

d. $y = \frac{1}{3}x - \frac{22}{3}$

Parallel lines share the same slope, $m = \frac{1}{3}$. $y - (8) = \frac{1}{3}(x - (2))$, so $y = \frac{1}{3}x + \frac{22}{3}$.

55 Solve $|3x + 18| = 19$, giving your answer as the ordered pair of solutions (smaller first).

a. $(-\frac{40}{3}, \frac{1}{3})$

b. $(-\frac{37}{3}, \frac{1}{3})$ ✓

c. $(-\frac{37}{3}, \frac{4}{3})$

d. $(\frac{1}{3}, -\frac{37}{3})$

Either $3x + 18 = 19$, giving $x = \frac{1}{3}$, or $3x + 18 = -19$, giving $x = -\frac{37}{3}$. Solutions: $(-\frac{37}{3}, \frac{1}{3})$.

56 Solve $|4x + 24| = 26$, giving your answer as the ordered pair of solutions (smaller first).

a. $(-\frac{27}{2}, \frac{1}{2})$

b. $(-\frac{25}{2}, \frac{3}{2})$

c. $(-\frac{25}{2}, \frac{1}{2})$ ✓

d. $(\frac{1}{2}, -\frac{25}{2})$

Either $4x + 24 = 26$, giving $x = \frac{1}{2}$, or $4x + 24 = -26$, giving $x = -\frac{25}{2}$. Solutions: $(-\frac{25}{2}, \frac{1}{2})$.

57 Solve $|8x - 48| = 15$, giving your answer as the ordered pair of solutions (smaller first).

a. $(\frac{33}{8}, \frac{71}{8})$

b. $(\frac{33}{8}, \frac{63}{8})$ ✓

c. $(\frac{63}{8}, \frac{33}{8})$

d. $(\frac{25}{8}, \frac{63}{8})$

Either $8x - 48 = 15$, giving $x = \frac{63}{8}$, or $8x - 48 = -15$, giving $x = \frac{33}{8}$. Solutions: $(\frac{33}{8}, \frac{63}{8})$.

58 Solve $|3x + 3| = 7$, giving your answer as the ordered pair of solutions (smaller first).

a. $(-\frac{10}{3}, \frac{4}{3})$ ✓

b. $(\frac{4}{3}, -\frac{10}{3})$

c. $(-\frac{10}{3}, \frac{7}{3})$

d. $(-\frac{13}{3}, \frac{4}{3})$

Either $3x + 3 = 7$, giving $x = \frac{4}{3}$, or $3x + 3 = -7$, giving $x = -\frac{10}{3}$. Solutions: $(-\frac{10}{3}, \frac{4}{3})$.

59 Solve $|8x - 64| = 3$, giving your answer as the ordered pair of solutions (smaller first).

a. $(\frac{67}{8}, \frac{61}{8})$

b. $(\frac{61}{8}, \frac{75}{8})$

c. $(\frac{61}{8}, \frac{67}{8})$ ✓

d. $(\frac{53}{8}, \frac{67}{8})$

Either $8x - 64 = 3$, giving $x = \frac{67}{8}$, or $8x - 64 = -3$, giving $x = \frac{61}{8}$. Solutions: $(\frac{61}{8}, \frac{67}{8})$.

60 A student has 127 dollars and wants to buy raffle entries costing 13 dollars each, keeping at least 10 dollars in reserve. Find the maximum number of raffle entries the student can buy.

a. 8

b. 9 ✓

c. 11

d. 10

$127 - 13n \geq 10$, so $-13n \geq -117$, giving $n \leq 9$ (dividing by a negative number flips the inequality). The student can buy at most 9 raffle entries.

- 61 A student has 27 dollars and wants to buy phone cases costing 4 dollars each, keeping at least 7 dollars in reserve. Find the maximum number of phone cases the student can buy.

a. 7
b. 4
c. 5 ✓
d. 6

$27 - 4n \geq 7$, so $-4n \geq -20$, giving $n \leq 5$ (dividing by a negative number flips the inequality). The student can buy at most 5 phone cases.

- 62 A student has 186 dollars and wants to buy concert tickets costing 12 dollars each, keeping at least 18 dollars in reserve. Find the maximum number of concert tickets the student can buy.

a. 16
b. 14 ✓
c. 15
d. 13

$186 - 12n \geq 18$, so $-12n \geq -168$, giving $n \leq 14$ (dividing by a negative number flips the inequality). The student can buy at most 14 concert tickets.

- 63 A student has 117 dollars and wants to buy raffle entries costing 13 dollars each, keeping at least 0 dollars in reserve. Find the maximum number of raffle entries the student can buy.

a. 11
b. 10
c. 8
d. 9 ✓

$117 - 13n \geq 0$, so $-13n \geq -117$, giving $n \leq 9$ (dividing by a negative number flips the inequality). The student can buy at most 9 raffle entries.

- 64 A student has 119 dollars and wants to buy concert tickets costing 5 dollars each, keeping at least 19 dollars in reserve. Find the maximum number of concert tickets the student can buy.
- a. 19
 - b. 22
 - c. 20 ✓
 - d. 21

$119 - 5n \geq 19$, so $-5n \geq -100$, giving $n \leq 20$ (dividing by a negative number flips the inequality). The student can buy at most 20 concert tickets.